GE-EHR for Monogenic Diabetes: Reclassification-Aware Genomic Decision Support in the Electronic Health Record

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Abstract—**Monogenic diabetes accounts for roughly 1–2% of diabetes, yet about 80% of cases are misdiagnosed as type 1 or type 2 diabetes, even though the causal gene directly determines treatment: GCK-related diabetes usually needs no medication, HNF1A- and HNF4A-related diabetes responds to low-dose sulfonylureas, and KCNJ11/ABCC8 neonatal diabetes responds to high-dose sulfonylureas. Many variants found in these genes are first reported as variants of uncertain significance (VUS), and their interpretation changes as evidence accumulates: re-curation with gene-specific expert-panel guidelines recently upgraded half of the GCK VUS in one clinical cohort to likely pathogenic. When an EHR evaluates a patient's variants only once, such reclassifications never reach the treating clinician, and a patient may remain on insulin when tablets would suffice. This paper presents the design of GE-EHR for monogenic diabetes, a Genomic-Enhanced EHR that stores VCF-derived variants in seven monogenic diabetes genes, classifies them against a versioned ClinVar knowledge base, applies six treatment-focused rules that consider the patient's current diabetes medications, and re-evaluates stored patients whenever ClinVar or the rules change. Upgrades into an actionable class trigger new alerts, downgrades supersede existing alerts for clinician review, and changes within a class update provenance silently. We position the design against existing reclassification systems and define an evaluation comparing an alert-once baseline, full recomputation and change-driven re-evaluation on real ClinVar release pairs and synthetic patients.**

Keywords—***Monogenic diabetes, MODY, neonatal diabetes, variant reclassification, clinical decision support, electronic health records, ClinVar, HL7 FHIR***

1. Introduction

Diabetes is usually classified as type 1 or type 2, but a minority of patients have *monogenic diabetes*, caused by a variant in a single gene. Monogenic diabetes makes up about 1–2% of diabetes cases, and approximately 80% of cases are misdiagnosed as type 1 or type 2 diabetes [1]. The diagnosis matters because treatment depends on the gene: no treatment in GCK diabetes, low-dose sulfonylurea in HNF1A/HNF4A diabetes, and high-dose sulfonylurea in neonatal diabetes caused by activating KCNJ11 or ABCC8 variants [1]. In a 2017–2024 cohort, 75% (18 of 24) of eligible patients underwent actionable treatment changes after their monogenic diagnosis was confirmed [2], and the ADA/EASD second international consensus report identifies monogenic diabetes as the area where precision medicine is already aiding diagnosis and treatment [3]. A genetic diagnosis also matters for families: because MODY is inherited in an autosomal dominant pattern, each first-degree relative has a 50% chance of carrying the same variant.

Electronic health records (EHRs) can turn these findings into decision support, but genomic decision support usually evaluates a patient's variants once. Variant interpretations change. When one group re-curated GCK VUS using the ClinGen gene-specific guidelines, four of eight were upgraded to likely pathogenic [4]. The ClinGen Monogenic Diabetes Expert Panel continues to publish gene-specific specifications and expert classifications to ClinVar [5]. Across genetic testing more broadly, a 2025 review found that at least 1.6% of variant classifications used in EHR care were outdated relative to ClinVar, including reclassifications that never reached the patient [6].

The clinical consequence in diabetes is concrete. A young patient labelled as having type 1 diabetes may

carry an HNF1A VUS and be treated with insulin. If that variant is later upgraded to likely pathogenic, the patient may be able to switch to low-dose sulfonylurea tablets—but only if someone notices. Existing reclassification tools address parts of this problem (Section 2.4), but none re-evaluates stored genotypes inside the EHR against the patient's current diabetes medications.

This paper's contributions are:

1. A versioned knowledge model for seven monogenic diabetes genes, in which every ClinVar release and rule revision is stored and every alert records the knowledge version behind it.
2. A change-driven re-evaluation algorithm that rechecks only stored variants whose classification changed, independent of the testing laboratory.
3. Six medication-aware treatment rules that compare the genetic diagnosis with the patient's current diabetes therapy, plus a family cascade-testing prompt.
4. An evaluation design comparing an alert-once baseline, full recomputation and change-driven re-evaluation on real ClinVar release pairs and synthetic patients.

2. Related Work

2.1 Monogenic Diabetes and Its Treatment

GCK-hyperglycaemia is the most prevalent form of monogenic diabetes [4]. Monogenic diabetes should be suspected for particular combinations of clinical and laboratory features, and accurate diagnosis changes treatment [1]: GCK diabetes causes mild, non-progressive fasting hyperglycaemia that usually requires no treatment, HNF1A/HNF4A diabetes is sulfonylurea-sensitive, and KCNJ11/ABCC8 neonatal diabetes responds to high-dose sulfonylurea [1]. A 2024 systematic review for the ADA/EASD Precision Medicine in Diabetes Initiative found randomized evidence that sulfonylureas are more effective in HNF1A-diabetes than in type 2 diabetes, but limited evidence for HNF4A-diabetes and, overall, limited evidence with moderate or serious risk of bias to guide monogenic diabetes treatment [7].

2.2 Variant Interpretation in Diabetes Genes

The ACMG/AMP guidelines define five classification tiers [8], and ClinVar aggregates submitted interpretations and retains previous versions of a submission when it is updated [9]. The ClinGen Monogenic Diabetes Variant Curation Expert Panel has published gene-specific specifications of these guidelines for GCK, HNF1A and HNF4A [5]. Applying them changes results: VUS re-curation using the GCK specification produced likely pathogenic classifications for half of the VUS reviewed [4].

2.3 Genomic Results in the EHR

Reclassifications usually reach clinicians only when a laboratory issues an amended report, a channel the 2025 Brotman Baty study found unreliable [6].

2.4 Existing Reclassification Systems

GeneInsight Clinic, developed by Partners HealthCare (now Mass General Brigham) and deployed from 2010, sends clinicians automated alerts when its laboratory changes the classification of a variant found in their patients, using high, medium and low alert levels [10]. Genome Alert! automatically reports clinically significant classification changes between ClinVar releases for laboratory reinterpretation; re-examining 4,929 sequencing files highlighted 45 changes and led to four additional diagnoses [11]. Most recently, automated reanalysis of genomic data at scale for rare-disease diagnosis has been shown to identify new findings, including variant downgrades, over time [12]. Laboratory systems cover only their own tests, and reanalysis tools work outside the prescribing context. Neither connects a reclassification in a diabetes gene to the patient's current diabetes treatment (Table I).

2.5 Genomic Interoperability

The HL7 Genomics Reporting Implementation Guide, version 2.0.0, defines FHIR profiles for reporting genomic variants [13].

Table I : Comparison with existing genomic decision-support and reclassification systems.

System	Knowledge source for updates	Where it acts	Considers current diabetes therapy	Re-applies revised treatment rules
GeneInsight Clinic [10]	Single laboratory reclassifications	Clinician portal linked to EHR	Not reported	Not reported
Genome Alert! [11]	ClinVar release changes	Laboratory reanalysis	No (reinterpretation tool)	Not reported
Automated reanalysis (Talos) [12]	Periodic reanalysis of genomic data	Rare-disease diagnostics	No (diagnostic tool)	Not reported
GE-EHR (this work)	**ClinVar releases (any laboratory), including expert-panel classifications**	**EHR decision support**	**Yes**	**Yes**

3. System Design

GE-EHR is a three-layer web application with five modules; the fifth, the re-evaluation engine, is the main contribution.

Table II : Functional requirements by module.

Module	Requirements
F1. Patient and genomic records	Patients, diabetes diagnosis and age at diagnosis, current diabetes medications; VCF upload; one patient view
F2. Variant alerts	Match variants in 7 diabetes genes to the current ClinVar release; ACMG/AMP labels; prefer expert-panel classifications
F3. Decision support	Six versioned, medication-aware rules with recommendation and source; advisory only
F4. FHIR export	Patient and variant Observations as FHIR R4, with amended status after reclassification
F5. Re-evaluation engine	Import new ClinVar releases and rule versions; detect changed classifications among stored variants; re-alert, supersede or update provenance; log every run

Non-functional requirements are role-based access for clinicians and lab technicians, hashed passwords, synthetic data only, and full provenance for every alert.

3.1 Architecture

F1–F3 run when a VCF is uploaded or a medication changes; F5 runs when a new ClinVar release or rule version is imported and re-runs F2/F3 for affected patients only (Figure I).

3.2 Versioned Data Model

Each ClinVar import is a `KB_RELEASE`; each `KNOWN_VARIANT` belongs to one release and records its review status. Rules carry a version and evidence source. Each `ALERT` stores the release and rule version that produced it and a status of *active*, *superseded* or *reviewed*; alerts are never deleted. The medication table records diabetes therapy (insulin, metformin, sulfonylurea, other), and the diagnosis table records diabetes type and age at diagnosis, which rule D4 needs.

4. Methods

4.1 VCF Parsing

VCF 4.2 files are parsed with cyvcf2. Each record becomes a variant key (chromosome, position, reference allele, alternate allele) on GRCh38 with zygosity; malformed files are rejected.

4.2 Gene Panel, Matching and Classification

The knowledge base holds ClinVar [9] records for seven monogenic diabetes genes: GCK, HNF1A, HNF4A, HNF1B, KCNJ11, ABCC8 and INS, with reference transcripts such as NM_000162.5 (GCK), NM_000545.6 (HNF1A) and NM_175914.4 (HNF4A). Variants are matched by exact key and take their ACMG/AMP category [8]; where a ClinGen expert-panel classification exists [5], it takes precedence. Pathogenic and likely pathogenic form the *actionable band*; VUS, likely benign, benign and not-in-

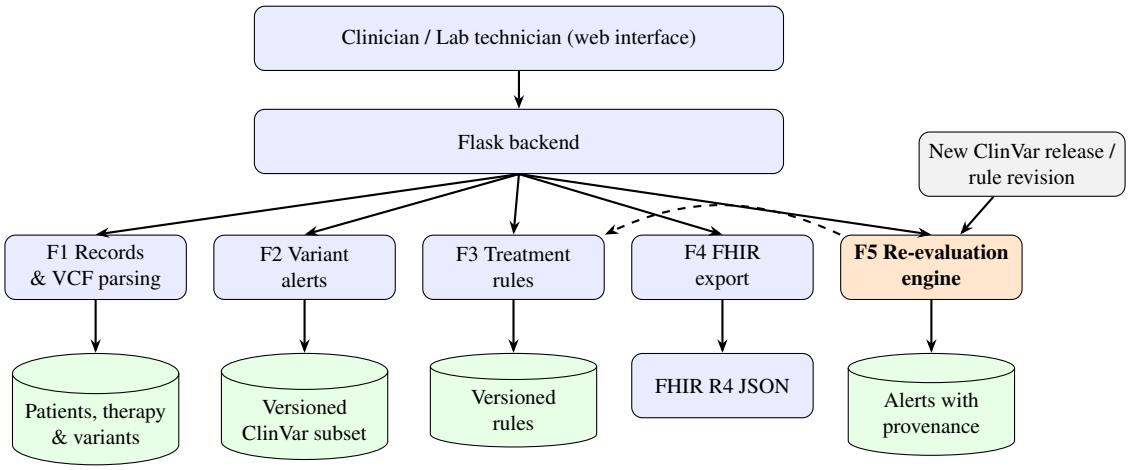


Fig I : GE-EHR architecture for monogenic diabetes. F5 is triggered by knowledge changes and re-runs F2/F3 (dashed arrow) for affected patients only.

knowledge-base form the *non-actionable band*. Unmatched variants are recorded as "not classified", not as VUS.

4.3 Medication-Aware Rules

Each rule has the form IF actionable variant in gene G (AND therapy condition T) THEN alert with severity S, recommendation R and source E (Table III, Algorithm 1). Recommendations are advisory and phrased for specialist review; GE-EHR never changes therapy itself.

Algorithm 1 Rule evaluation for one patient

Require: patient p, current release K, rule set $\mathscr{R}$
$V \leftarrow$ classified variants of p under K
$T \leftarrow$ current diabetes therapy and diagnosis data of p
for each rule $r \in \mathscr{R}$ **do**
  if $r.genetic \in V$ **then**
    if $r.therapy = \varnothing$ **or** $r.therapy$ holds for T **then**
      create alert(p, r, K) with provenance
    else
      add info note to genomic profile of p
    end if
  end if
end for

4.4 Reclassification-Aware Re-evaluation

When a new ClinVar release is imported, F5 compares it with the current release only for variant keys stored for some patient. Each changed variant is assigned a category (Table IV) and Algorithm 2 applies the action to every carrier. Because monogenic diabetes is inherited, a single upgraded variant often affects several related patients, all of whom are re-evaluated.

Algorithm 2 Change-driven re-evaluation

Require: releases K_{old}, K_{new}; stored variant keys V_{obs}
$\Delta \leftarrow \{v \in V_{obs} : class_{K_{old}}(v) \neq class_{K_{new}}(v)\}$
for each $v \in \Delta$ **do**
  $c \leftarrow category(class_{K_{old}}(v), class_{K_{new}}(v))$
  for each patient p carrying v **do**
    if $c = upgrade$ **then**
      run Algorithm 1 for p under K_{new}; tag *reclassified*
    else if $c = downgrade$ **then**
      mark alerts of p from v *superseded*; issue review notice
    else
      update provenance of p's alerts to K_{new}
    end if
  end for
end for
set K_{new} as current; log run

Work scales with the number of changed variants and their carriers, not with the size of ClinVar. Downgrades never silently delete an alert or reverse treatment: a patient may already have switched therapy, so a clinician must review the change.

Table III : Monogenic diabetes decision-support rules (current versions).

Rule	Genetic condition	Therapy condition	Severity	Recommendation	Source
D1	GCK pathogenic / likely pathogenic	Any glucose-lowering drug	Medium	Medication usually not required outside pregnancy; review whether it can be stopped	[1, 4]
D2	HNF1A pathogenic / likely pathogenic	Insulin or metformin, no sulfonylurea	High	Consider low-dose sulfonylurea as first-line treatment	[1, 7]
D3	HNF4A pathogenic / likely pathogenic	Insulin or metformin, no sulfonylurea	Medium	Consider low-dose sulfonylurea (supporting evidence is more limited than for HNF1A)	[1, 7]
D4	KCNJ11 or ABCC8 activating pathogenic / likely pathogenic, diabetes diagnosed in infancy	Insulin	High	Refer to specialist to consider transfer to high-dose sulfonylurea	[1]
D5	Any of the 7 genes pathogenic / likely pathogenic	—	Low	Offer genetic counselling and cascade testing of first-degree relatives (50% inheritance)	Autosomal dominant inheritance
D6	GCK, HNF1A or HNF4A VUS	—	Info	No treatment change; flag as monitored for reclassification	[5, 8]

Table IV : Change categories and system actions.

Change	Example	Action
Upgrade into actionable band	HNF1A VUS → LP	Run rules; new alert marked *reclassified*, high priority
Downgrade out of actionable band	GCK LP → VUS	Supersede alert; review notice; FHIR status *amended*; never reverse therapy automatically
Within actionable band	HNF4A LP → P	Update provenance; no notification
Within non-actionable band	VUS → LB	Update provenance; no notification
Rule revision	New guidance	Re-run revised rule for carriers of its genes; notify only if outcome changes

4.5 FHIR Export and Implementation

`GET /patients/{id}/fhir` returns a FHIR R4 Bundle using a simplified subset of the HL7 Genomics Reporting variant profile [13]; reclassified Observations are exported with status *amended*. GE-EHR is designed for implementation in Python with Flask, SQLAlchemy over SQLite, cyvcf2 and fhir.resources, with a scheduled release-import job and pytest unit tests.

5. Evaluation

5.1 Data

Patients. A correctness cohort of 30 synthetic patients and a scale cohort of 10,000 are generated with Synthea [14], using its diabetes module for diagnoses and medications. Each patient receives a VCF with selected ClinVar variants in the seven genes plus benign background variants. Family groups are created so that one variant is shared by several related patients.

Knowledge changes. A pair of real archived ClinVar releases several years apart, restricted to the seven genes, provides natural reclassifications; a controlled change set injects known upgrades, downgrades and within-band changes, plus a simulated revision of rule D4 (version 1 covers KCNJ11 only; version 2 adds ABCC8).

5.2 Strategies Compared

(1) **Alert-once baseline**: alerts generated at upload or prescribing and never re-evaluated; (2) **full recomputation** of every patient on each release; (3) **change-driven re-evaluation** (GE-EHR).

Table V : Test scenarios.

ID	Scenario	Expected
S1	HNF1A pathogenic, on insulin	High alert, D2
S2	HNF1A pathogenic, already on sulfonylurea	Info note only
S3	GCK pathogenic, on metformin	Medium alert, D1
S4	GCK pathogenic, no medication	Info note only
S5	HNF4A likely pathogenic, on metformin	Medium alert, D3
S6	KCNJ11 pathogenic, diagnosed in infancy, on insulin	High alert, D4
S7	Any actionable variant	Low alert, D5
S8	HNF1A VUS	Info, D6
S9	Only benign variants	No alert
S10	Malformed VCF	Rejected
S11	FHIR export of S1 patient	Valid FHIR R4 Bundle
S12	Lab technician opens recommendations	Access denied
RS1	HNF1A VUS (S8) upgraded to LP; patient on insulin	New high alert D2, *reclassified*
RS2	GCK LP downgraded to VUS	Alert superseded; review notice
RS3	HNF4A LP changed to P	Provenance updated; no notice
RS4	HNF1A VUS changed to LB	Provenance updated; no notice
RS5	D4 v1 (KCNJ11) replaced by v2 (adds ABCC8)	ABCC8 carriers on insulin newly alerted
RS6	Upgraded variant shared by a family of patients	Every carrier alerted
RS7	Changed variant carried by no patient	No work performed

5.3 Test Scenarios

5.4 Metrics and Success Criteria

Metrics are re-alert sensitivity (carriers correctly alerted after an upgrade / all carriers of upgraded variants), stale alerts remaining after a release, notification precision, records examined and time per release on the 10,000-patient cohort, and point-in-time correctness on S1–S12. Table VI states expected outcomes; measured values will be reported once the prototype is implemented.

Table VI : Success criteria by strategy (expected outcomes).

Metric	Alert-once	Full	GE-EHR
Re-alert sensitivity	0%	100%	100%
Stale alerts after release	All affected	0	0
Notification precision	n/a	100%	100%
Records examined	0	All	Changed only
Time per release (10k patients)	—	Baseline	Lower than full
Point-in-time correctness	100%	100%	100%

6. Discussion

Why diabetes is a strong use case. In many genetic conditions a reclassification changes risk counselling; in monogenic diabetes it can change daily treatment. An upgraded HNF1A or HNF4A variant may allow a patient to move from insulin injections to low-dose tablets, and an upgraded GCK variant may allow medication to be stopped [1]. Because VUS are common in these genes and active expert-panel curation [4, 5] makes reclassification likely, alert-once decision support leaves real benefit unrealised.

Relation to existing systems. GeneInsight showed that automated reclassification alerts are feasible [10], and Genome Alert! showed that ClinVar release comparison surfaces clinically significant changes [11]. GE-EHR adds the link to the patient's current diabetes therapy, so the output is a treatment-relevant alert rather than a reinterpretation report.

Families and alert burden. Monogenic diabetes is inherited, so one upgrade can benefit several relatives at once, and D5 prompts cascade testing. Most reclassifications that stay within the non-actionable band produce no notification, keeping alert burden low.

6.1 Limitations

- Synthetic patients only; clinical accuracy and clinician response are not evaluated.
- ClinVar may contain conflicting submissions;

GE-EHR prefers expert-panel classifications and prompts review rather than changing therapy.

- Seven genes and six rules; rarer monogenic and syndromic forms are not covered, and treatment in pregnancy needs specialist guidance beyond these rules.
- Treatment rules are simplified summaries of published guidance; the underlying treatment evidence is limited and at moderate or serious risk of bias [7], so rules must be reviewed by diabetes specialists before any clinical use.
- Exact variant matching; SQLite is not suited to heavy concurrent use.

6.2 Future Work

- Implementation of the prototype and reporting of measured results.
- Analysis of real ClinVar history for the seven genes, measuring reclassification rates and directions per gene.
- Integration of a MODY probability calculator to prompt genetic testing in patients with atypical type 1 or type 2 diabetes.
- Ingestion of laboratory amended reports through FHIR and full alignment with the HL7 Genomics Reporting profiles [13].

7. Conclusion

In monogenic diabetes, the gene determines the treatment, and the interpretation of that gene's variants changes over time. GE-EHR for monogenic diabetes is designed so that these changes reach the clinician: it versions the knowledge base and rules, records the evidence behind every alert, re-evaluates only the stored variants whose interpretation changed, and connects each change to the patient's current diabetes therapy. The design offers a reproducible way to make sure a patient who could move from insulin to tablets, or stop medication altogether, is not missed because the science moved on after their test.

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